

Answer: A

15 Total energy($E_k + \text{GPE} + \text{EPE}$) must be conserved.

$$\text{Maximum } E_k = \frac{1}{2} m (v_{\max})^2 = \frac{1}{2} (2.5)(1.2)^2 = 1.8J$$

Answer: A

16 First law of thermodynamics: $\Delta U = Q + W$